

Task 1: News Topic Classifier Using BERT

Objective:

Fine-tune a transformer model (e.g., BERT) to classify news headlines into topic categories.

Dataset:

AG News Dataset (Available on Hugging Face Datasets)

Instructions:

- Tokenize and preprocess the dataset
- Fine-tune the bert-base-uncased model using Hugging Face Transformers
- Evaluate the model using accuracy and F1-score
- Deploy the model using Streamlit or Gradio for live interaction

Skills Gained:

- NLP using Transformers
- Transfer learning & fine-tuning
- Evaluation metrics for text classification
- Lightweight model deployment